LAB EXPERIMENT RECORD

24-10-2026---Set 29: Student Academic Performance

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Register Number	192325017
Date	24 OCTOBER 2026
Day	SATURDAY

Aim

To analyze student academic data in R and Tableau using a stacked area chart, boxplot, density plot and an interactive scatter plot.

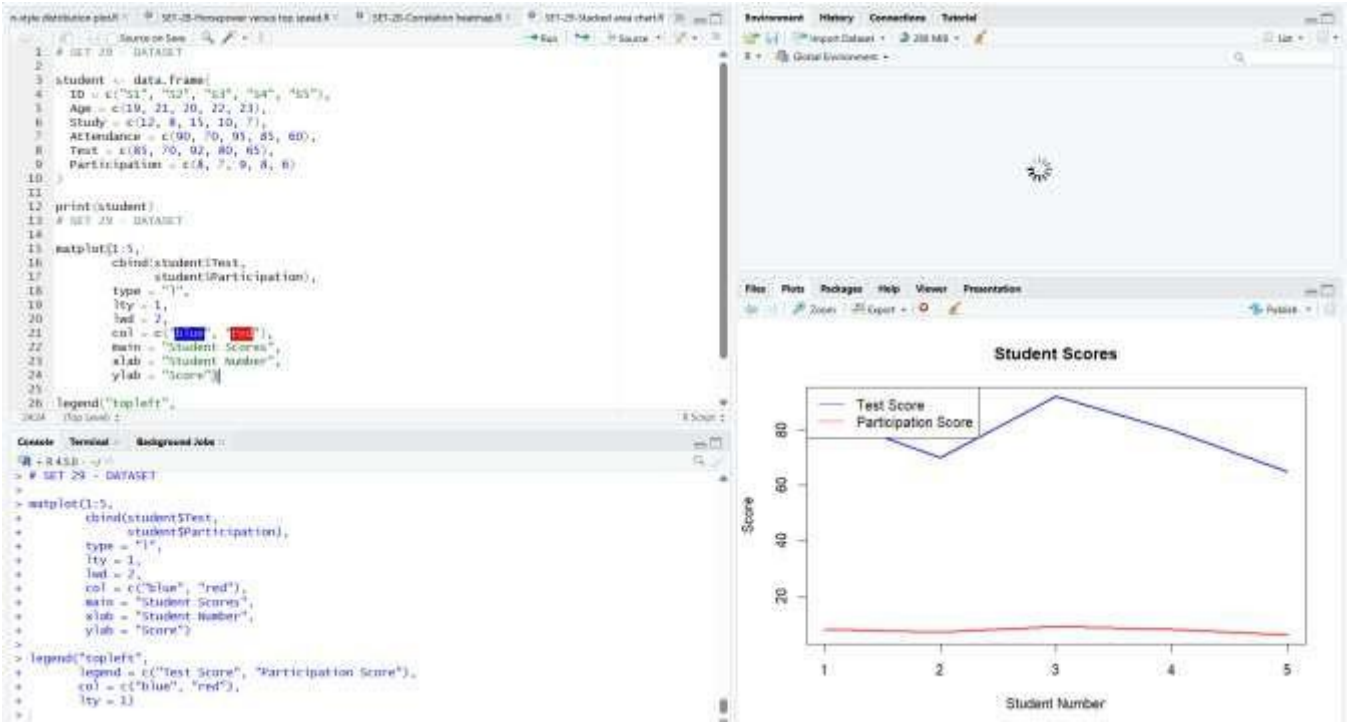
Dataset: Student Academic Performance

Student ID	Age	Study Hours	Attendance (%)	Test Score	Participation Score
S1	19	12	90	85	8
S2	21	8	70	70	7
S3	20	15	95	92	9
S4	22	10	85	80	8
S5	23	7	60	65	6

Question 1: Stacked Area Chart of Test and Participation Scores

Question: Using R, create a stacked area chart showing Test_Score and Participation_Score across students. Analyze trends.

Aim: To show how test scores and participation scores of each student add up and change across students using a stacked area chart.

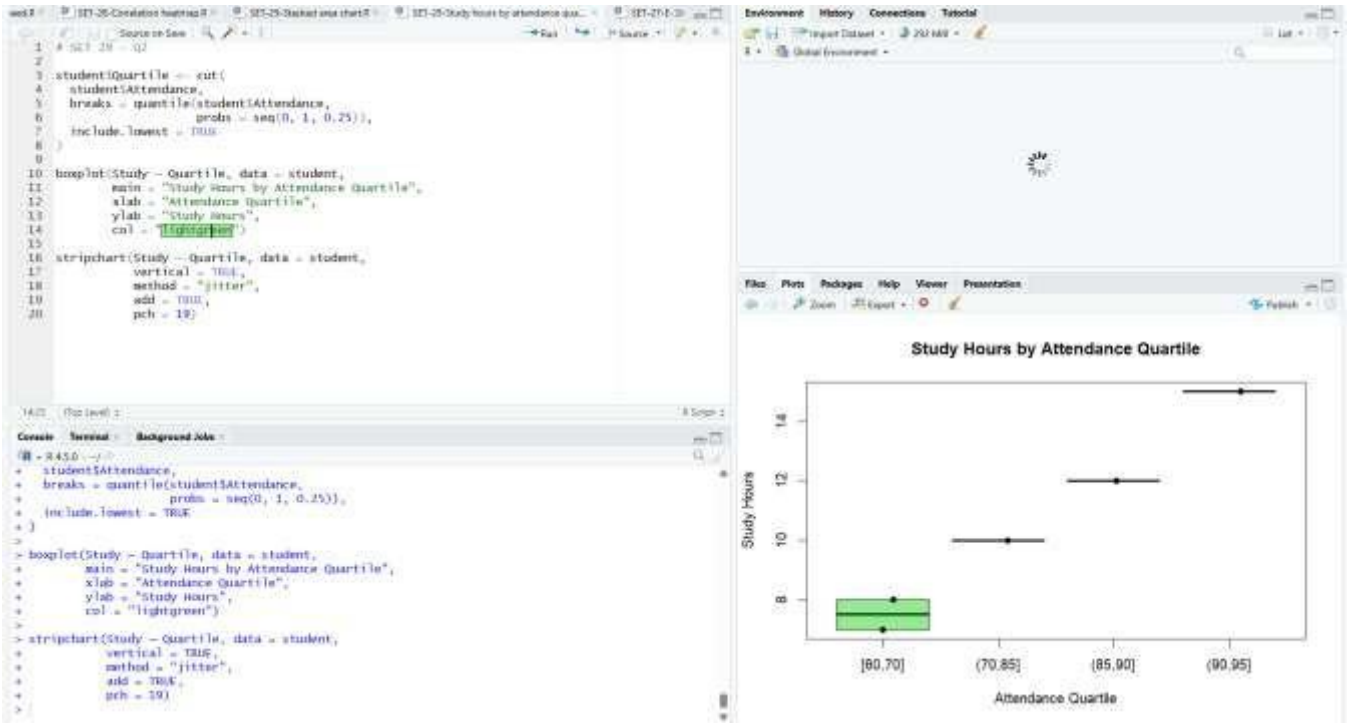


Result: Both scores rise and fall together: S3 is highest (92 and 9) and S5 is lowest (65 and 6); the participation layer is very thin because it is out of 10 while the test score is out of 100.

Question 2: Boxplot of Study Hours by Attendance Quartile

Question: Using R, generate a boxplot of Study_Hours grouped by Attendance quartiles. Interpret patterns. Include axis labels, title, and colors for each quartile. Analyze the boxplot to identify median and interquartile range of study hours per attendance quartile and any outliers in study hours.

Aim: To compare the study hours of students in the four attendance quartile groups using a coloured boxplot.

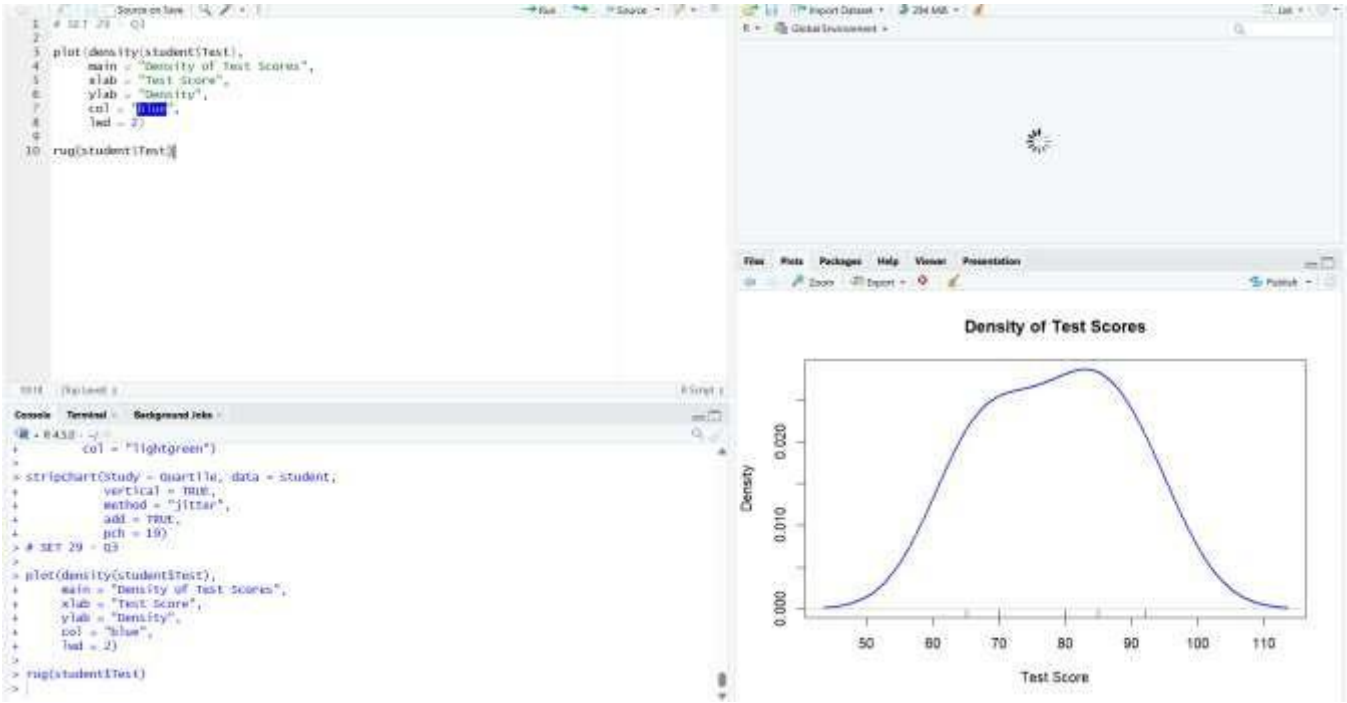


Result: The median study hours rise with attendance, from 7.5 in the lowest quartile to 10, 12 and 15 in the highest, the interquartile range is 4 hours overall (8 to 12), and there are no outliers.

Question 3: Density Plot of Test Score

Question: Using R, construct a density plot for Test_Score to evaluate score distribution and outliers.

Aim: To evaluate the shape of the test score distribution and look for outliers using a density plot.

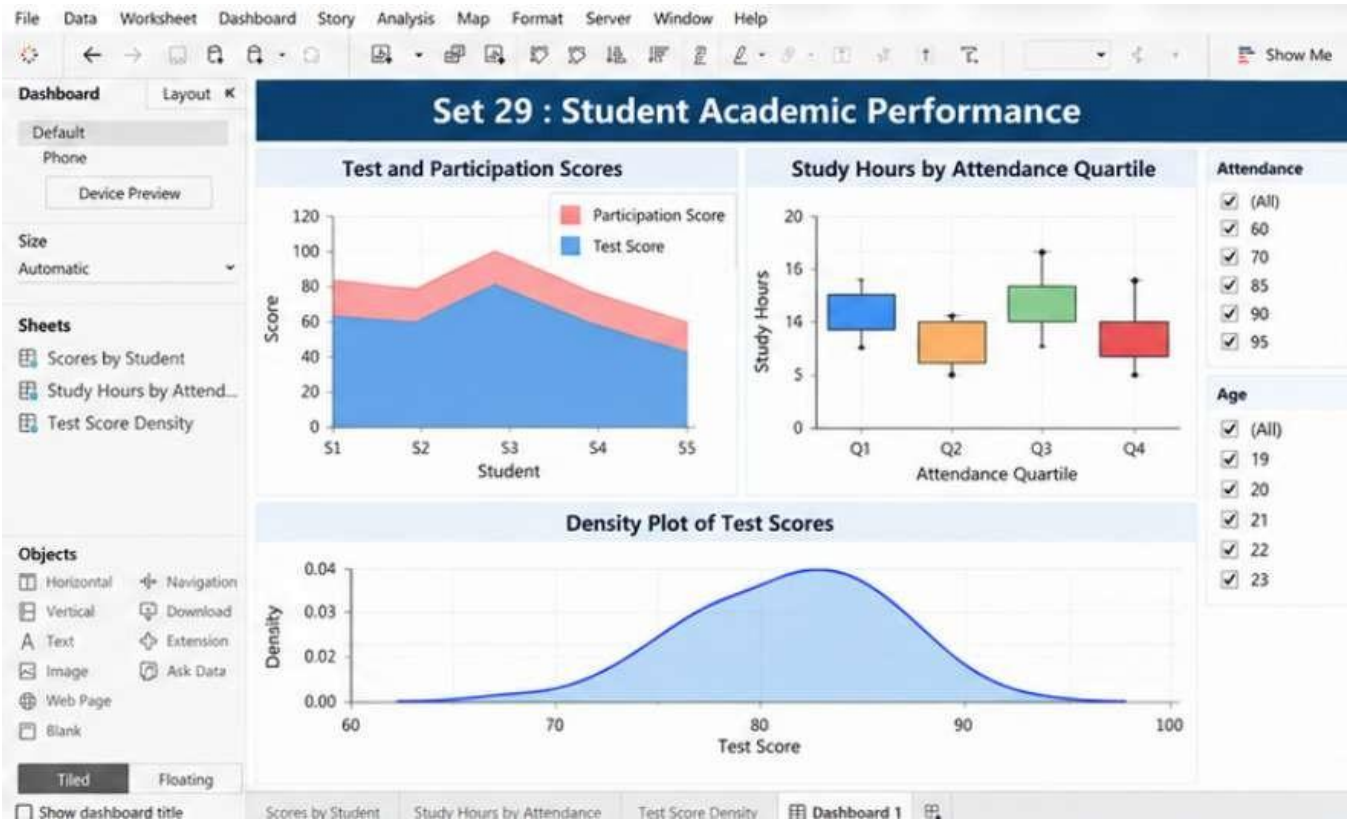


Result: Scores range from 65 to 92 with a mean of 78.4 and a median of 80, the curve is spread out and roughly symmetric, and there are no outliers.

Question 4: Interactive Tableau Scatter Plot

Question: In Tableau, design an interactive scatterplot with Study_Hours on X-axis, Test_Score on Y-axis, colored by Participation_Score. Include filters for Age.

Aim: To build an interactive Tableau scatter plot of study hours and test score, coloured by participation score and filtered by age.



Result: Test scores rise with study hours (correlation about 0.98), the darker high-participation points sit at the top right, and the Age filter shows that the older students (22 and 23) have lower scores.

Overall Result

Observation: The average test score is 78.4, average study time is 10.4 hours, average attendance is 80% and average participation is 7.6, and students who study and attend more score higher.